

Andy Huang

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EDUCATION

Rensselaer Polytechnic Institute

Expected May 2026

B.S. Computer Systems Engineering

- GPA: 3.24
- Courses: Microprocessor Systems, Computer Comm Networks, Computer Architecture and Operating Systems

EXPERIENCE

Full Stack Development Intern | HARP Research Inc.

May 2025 - August 2025

- Developing an internal productivity bot to support automated time tracking and Agile workflow for remote teams.
- Refactored hour-tracking queries into OOP-based SQLAlchemy logic; reduced code duplication by 40%
- Deployed the company website to production, performing database schema migration and backend integration.
- Developed unit tests for hour-tracking functions; cut runtime errors and debugging time by 30%

Undergraduate Research Student | Rensselaer Polytechnic Institute

January 2024 – September 2025

- Spearheaded development of Music X+, a hybrid music and information hub to improve student engagement and lab workflow efficiency.
- Built a web interface with Flask and HTML/CSS/JS to manage song requests, display attendance, and control music playback.
- Logged student attendance metadata from RFID scans into a MySQL database for accurate records and reporting
- Developed a Python backend leveraging the Genius API to enable real-time music search and playlist creation.
- Implemented UDP-based inter-computer communication to transmit staff info across the lab network.

PROJECTS

Telephone Drawing Game | C, STM32, Bare-Metal, UART, DMA

- Designed a dual-device drawing game using STM32 boards with LCD touchscreen input and SD card storage.
- Implemented JPEG encode/decode pipeline to save drawings using FatFs, DMA2D, and HAL JPEG drivers.
- Designed an interrupt-based UART file transfer protocol with explicit handshaking for image transmission.
- Implemented low-level LCD framebuffer rendering and touchscreen input handling using STM32 BSP.
- Structured application logic using a deterministic state machine to coordinate I/O and communication.

OLED Weather Display | C, FreeRTOS, ESP32, I2C

- Designed and implemented a real-time weather display on the ESP32 using FreeRTOS for low-latency updates.
- Synced time via SNTP and fetched weather data over HTTP using WiFi, ensuring accurate real-time information.
- Configured ESP-IDF HAL and registers for peripheral setup and precise control of I2C and display interfaces.
- Integrated I2C OLED display with LVGL for graphical UI rendering on the microcontroller.
- Managed concurrent tasks with FreeRTOS queues to coordinate display updates and network operations.

SMART Fire Alarm System | C, Raspberry Pi, ESP32, MQTT, AWS IoT

- Designed a WiFi system to transmit sensor data to the cloud, providing reliable alerts for at-risk users.
- Configured a Raspberry Pi to collect sensor data, trigger alarms, and sync with AWS IoT for cloud monitoring.
- Linked an ESP32 to the Raspberry Pi via MQTT to drive a remote vibrational alarm for hearing-impaired users.

TECHNICAL SKILLS

Languages: C, C++, Python, HTML, CSS, JavaScript, SQL, MIPS & ARM Assembly

Tools & Frameworks:: ESP-IDF, FreeRTOS, STM32 HAL, Docker, Flask, Django, Linux, Git, MySQL, PostgreSQL

Interfaces & Protocols:: UART, I2C, SPI, DMA, BLE, Wi-Fi (TCP/IP, UDP, SNTP, HTTP), MQTT

Embedded Platforms: ESP32, STM32F7 (Discovery), Raspberry Pi, Arduino

COMMUNITY & LEADERSHIP

House Manager | Delta Phi Fraternity

November 2023 – Present

- Directed day-to-day operations for a 25-member fraternity house, overseeing daily and weekly cleaning, and facility upkeep throughout the semester.
- Coordinated alumni, contractors, and service personnel, ensuring EHS compliance and cost-effective maintenance.